

### Data and knowledge layer

- **Neo4j:** Graph database used to store complex relationships between data points, useful for 'GraphRAG'.
- **Postgres:** Traditional relational database for structured data storage and transactional integrity.
- **ElasticSearch:** Distributed search engine used for rapid full-text indexing and retrieval.
- **Milvus:** Platform for highly scalable vector database for managing large-scale embedding data.

### Integration layer

- **Temporal:** Framework that ensures durable, reliable execution of long-running workflows, handling retries and state management.
- **Airbyte:** ETL/ELT platform with a library of connectors to sync data from various sources into the agent's knowledge layer.
- **GraphQL Federation:** Framework that unifies multiple microservices into a single graph, allowing agents to query across different services seamlessly.
- **REST/gRPC endpoints:** Standardised communication protocols for interoperability between the agent and other software services.

### Infrastructure layer

- **Kubernetes (K8s):** Framework that manages the deployment, scaling, and orchestration of agent containers.
- **Ray:** Framework for scaling AI and Python workloads, such as distributed training or parallelising agentic tasks.
- **Docker:** Provides the container runtime to ensure the agent's environment is consistent across different machines.
- **Terraform:** Enables infrastructure as code (IaC) to automatically provision the cloud resources needed for the AI stack.

### Operations and observability layer

- **Arize AI:** Platform that provides observability specifically for LLMs, tracking traces and identifying where a reasoning chain may have failed.
- **DeepEval:** Framework for testing and evaluating LLM outputs based on specific metrics.

- **RAGAS:** Designed to evaluate the performance of retrieval-augmented generation pipelines.
- **Prometheus:** Monitoring tool that collects and stores metrics (CPU, memory, latency) as time-series data.
- **Grafana:** Visualisation platform used to create dashboards for the metrics collected by Prometheus.

### Governance, security, and risk layer

- **Vault (HashiCorp):** Framework that manages sensitive information such as API keys, passwords, and certificates, ensuring agents access them securely.

These tools form a cohesive ecosystem that enables agents to operate with autonomy while remaining aligned with enterprise values and regulatory expectations.

To adopt agentic AI effectively, enterprises must first understand the fundamentals of agentic loops and develop a unified data and AI strategy that supports orchestration, tool integration, memory, and governance.

Agentic AI is not a competitor to human capability but a strategic enabler augmenting expertise, amplifying productivity, and powering the next generation of intelligent and autonomous enterprise operations. **END** 

### Acknowledgements

The authors would like to thank Tricon Solutions LLC and Gspann Technologies, Inc., for giving the required time and support in many ways when this article was being written.

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